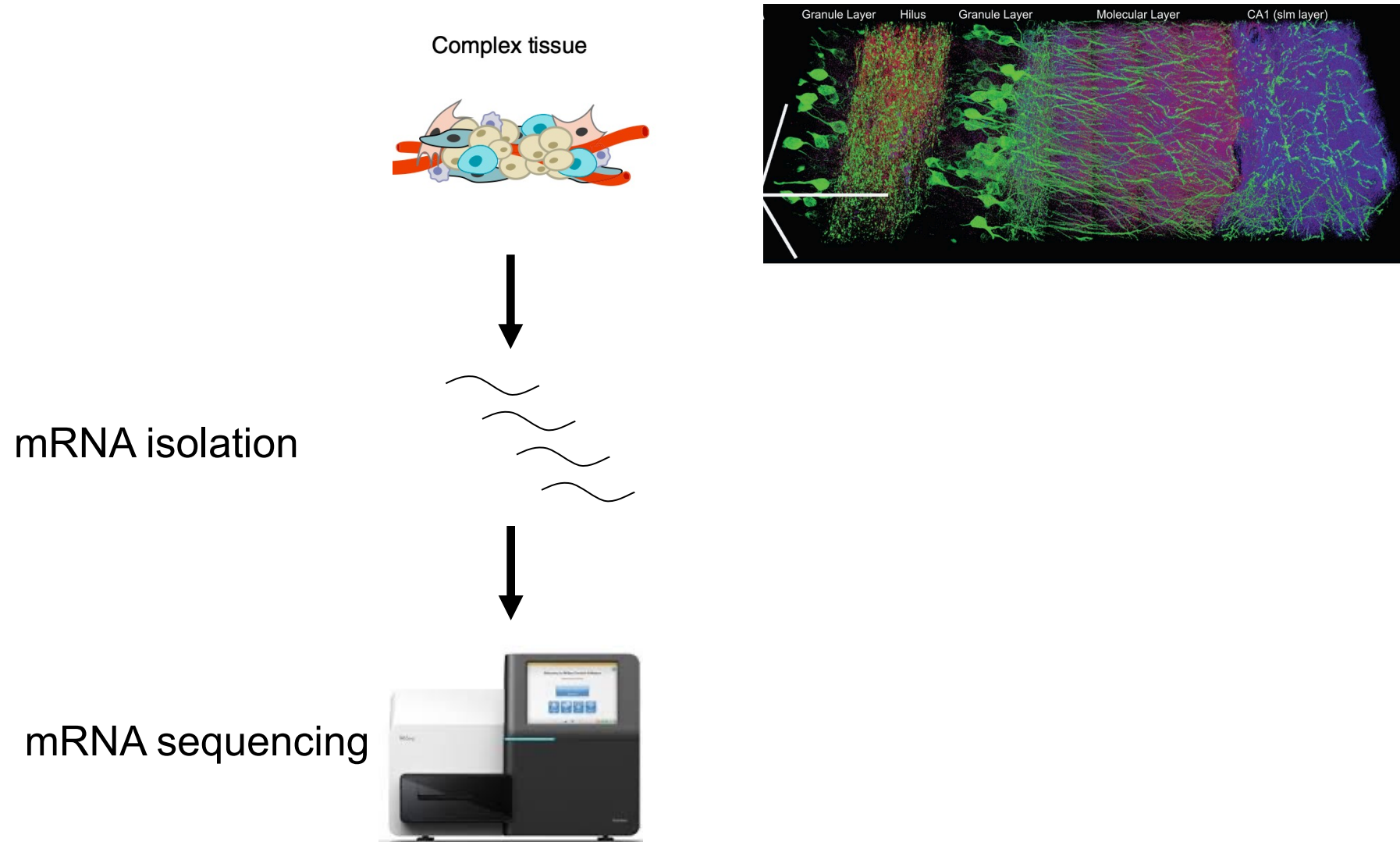


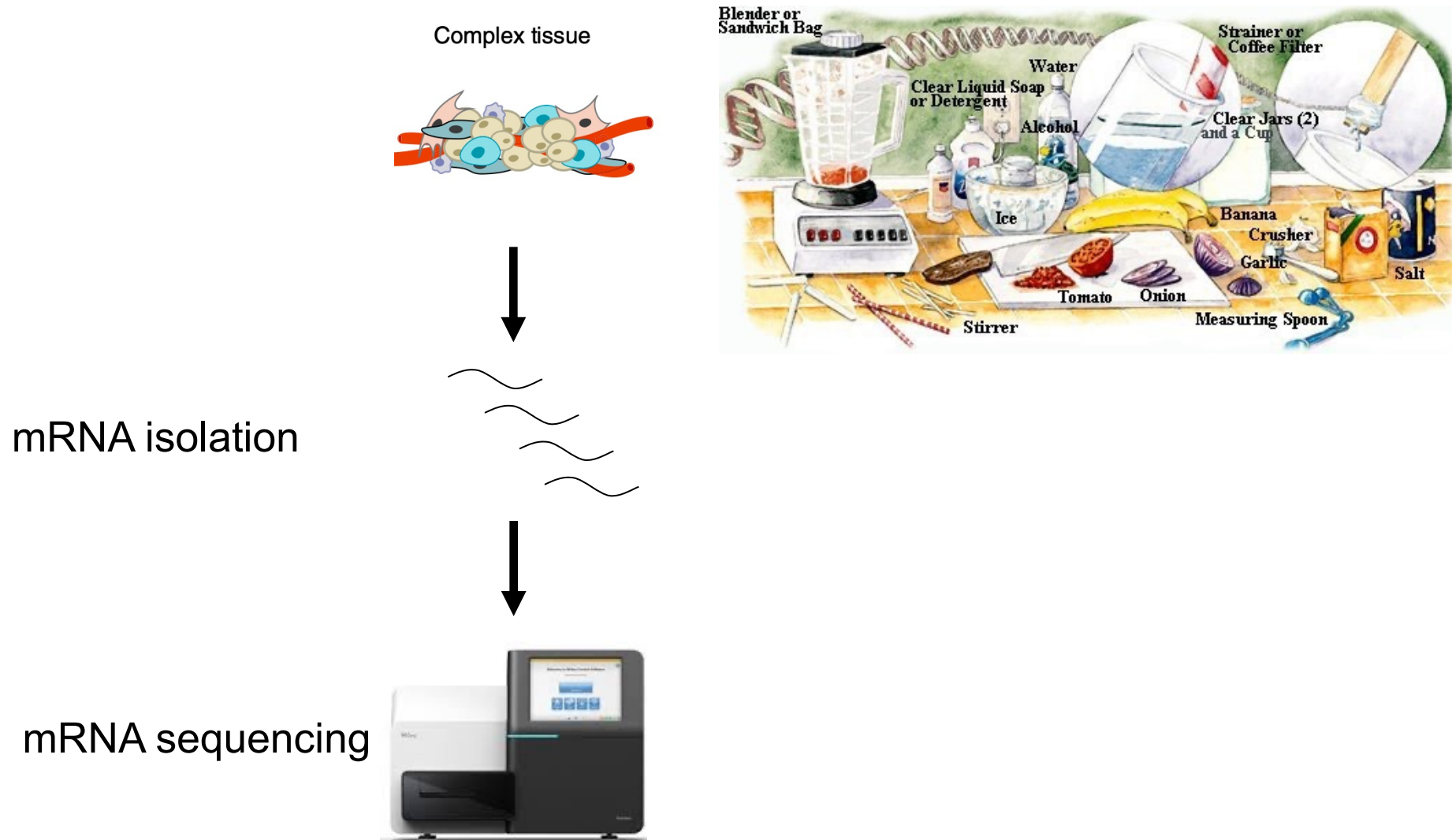
Single cell sequencing

Neuro-genomics
Class 8

What do we lose in bulk sequencing?

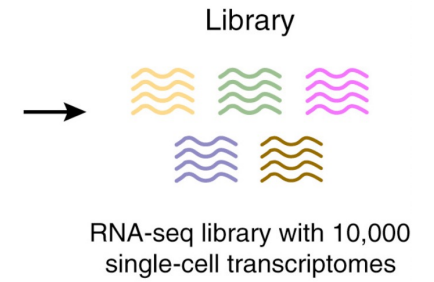


What do we lose in bulk sequencing?



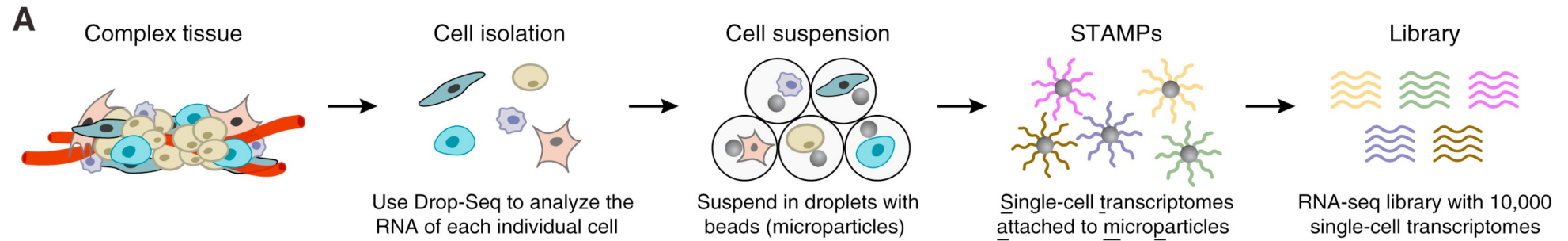
Single cell RNA sequencing

A Complex tissue



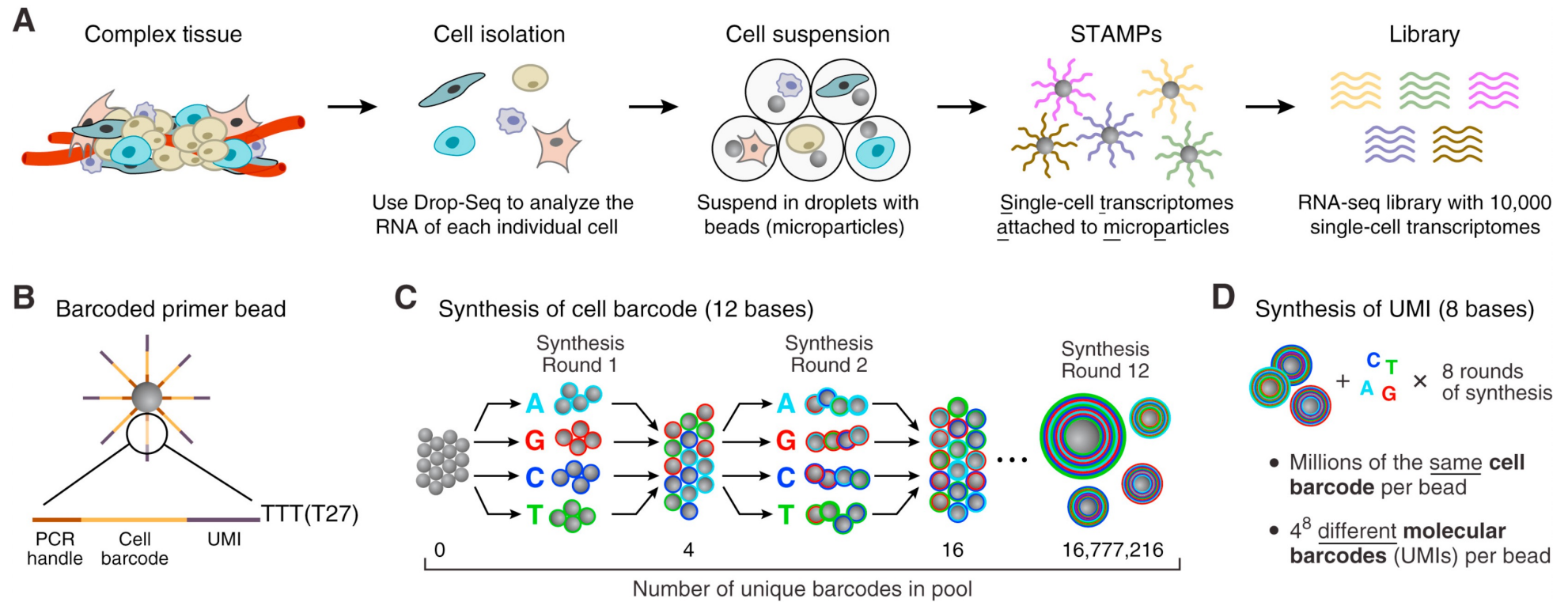
Macosko et al., Cell, 2015

Single cell RNA sequencing



Macosko et al., Cell, 2015

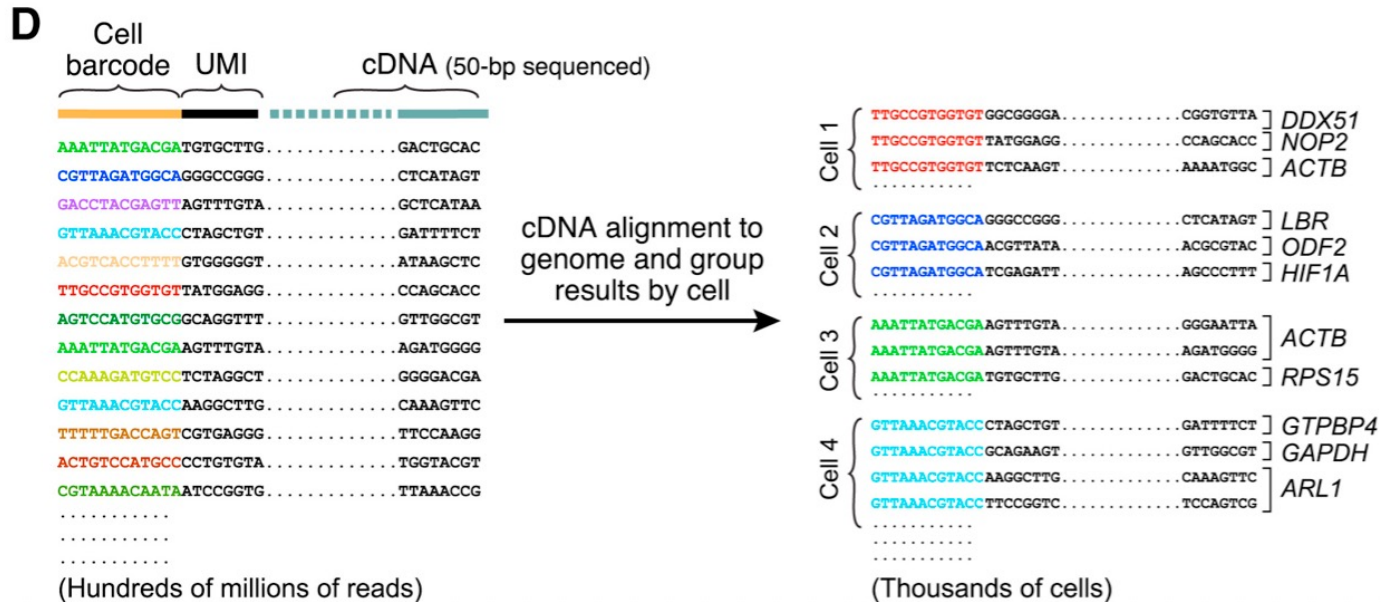
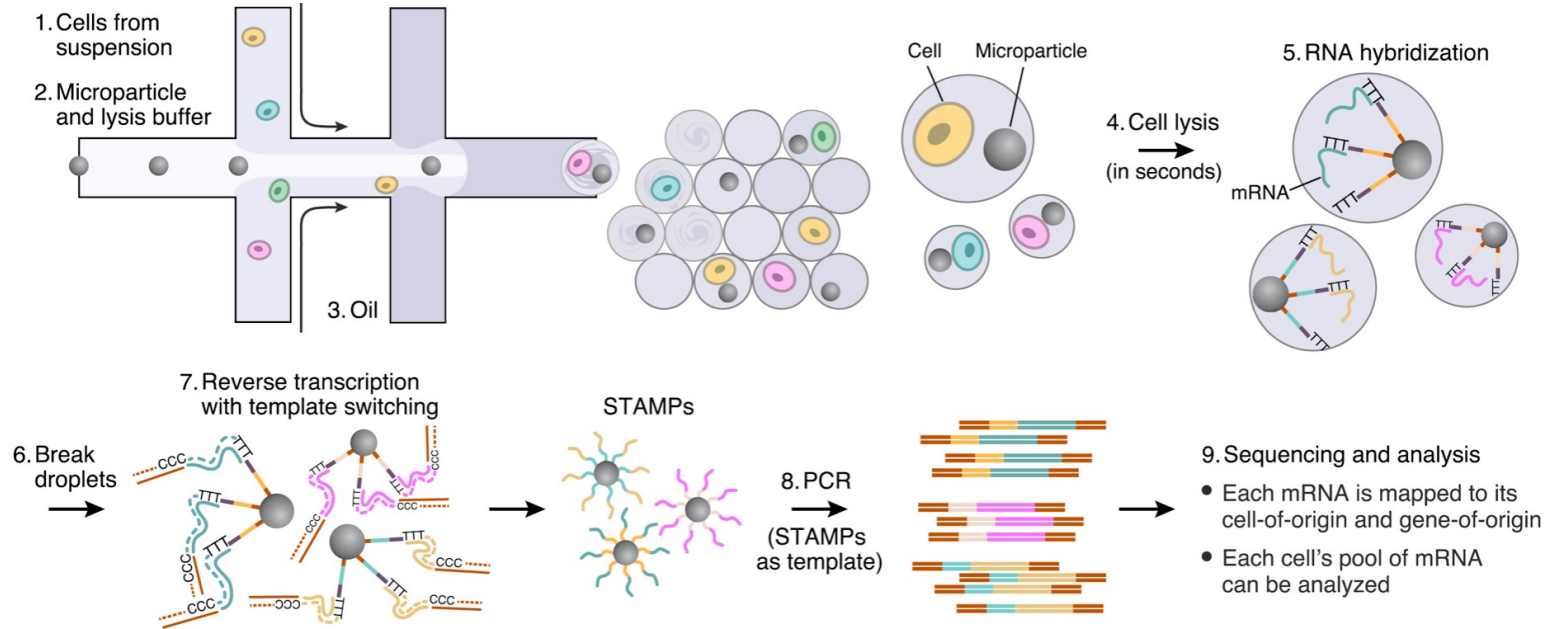
Single cell RNA sequencing



Macosko et al., Cell, 2015

- Note that two barcodes are used: unique molecular identifier (UMI) and unique cell barcode

Single cell RNA sequencing



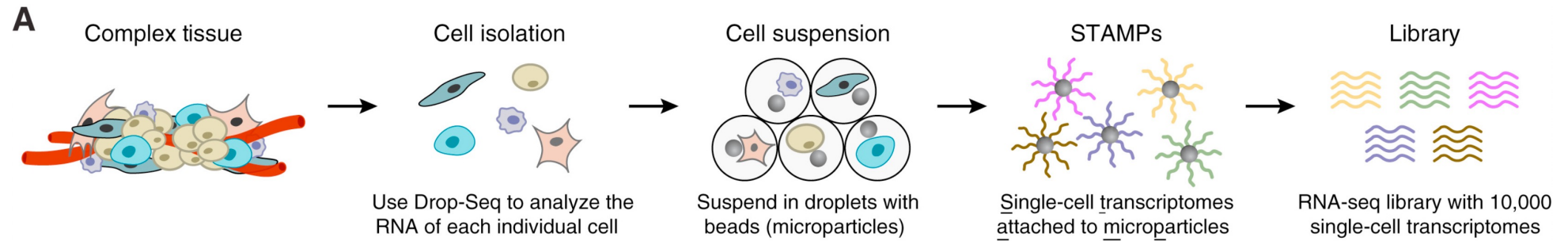
Count unique UMIs for each gene in each cell

→

Create digital expression matrix

Cell:		1	2	...	N
GENE 1		1	2		14
GENE 2		4	27		8
GENE 3		0	0		1
...		...	...		...
GENE M		6	2		0

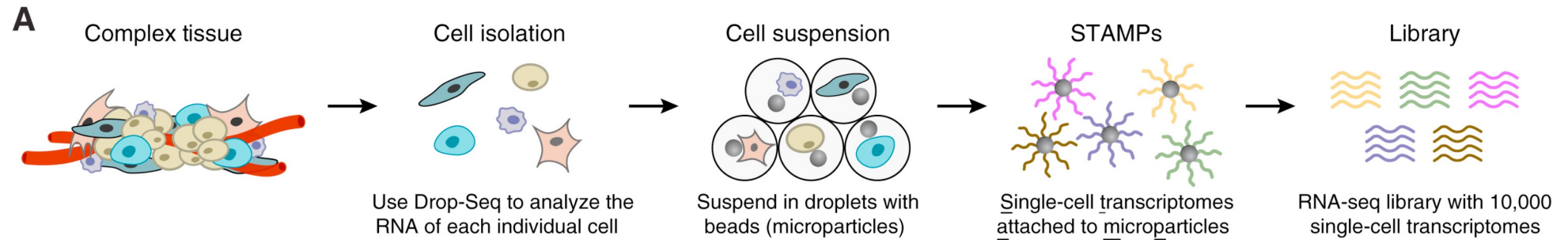
Single cell RNA sequencing



Macosko et al., Cell, 2015

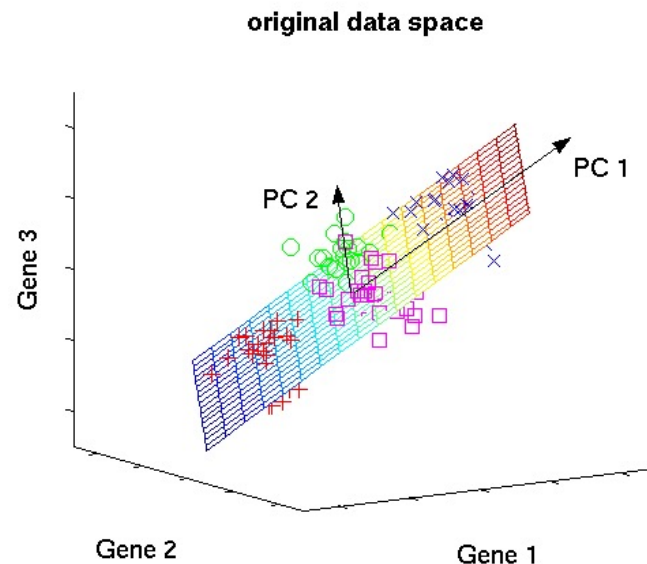
- How can we compare cells when we have ~20,000 dimensions?

Single cell RNA sequencing



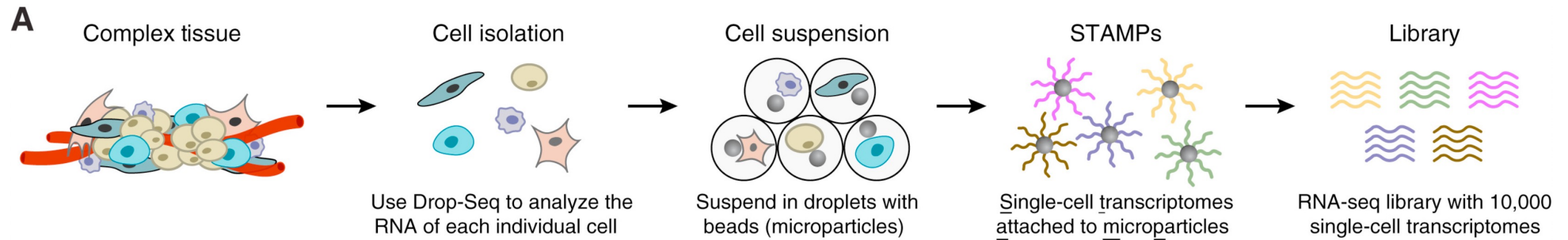
Macosko et al., Cell, 2015

- How can we compare cells when we have ~20,000 dimensions?



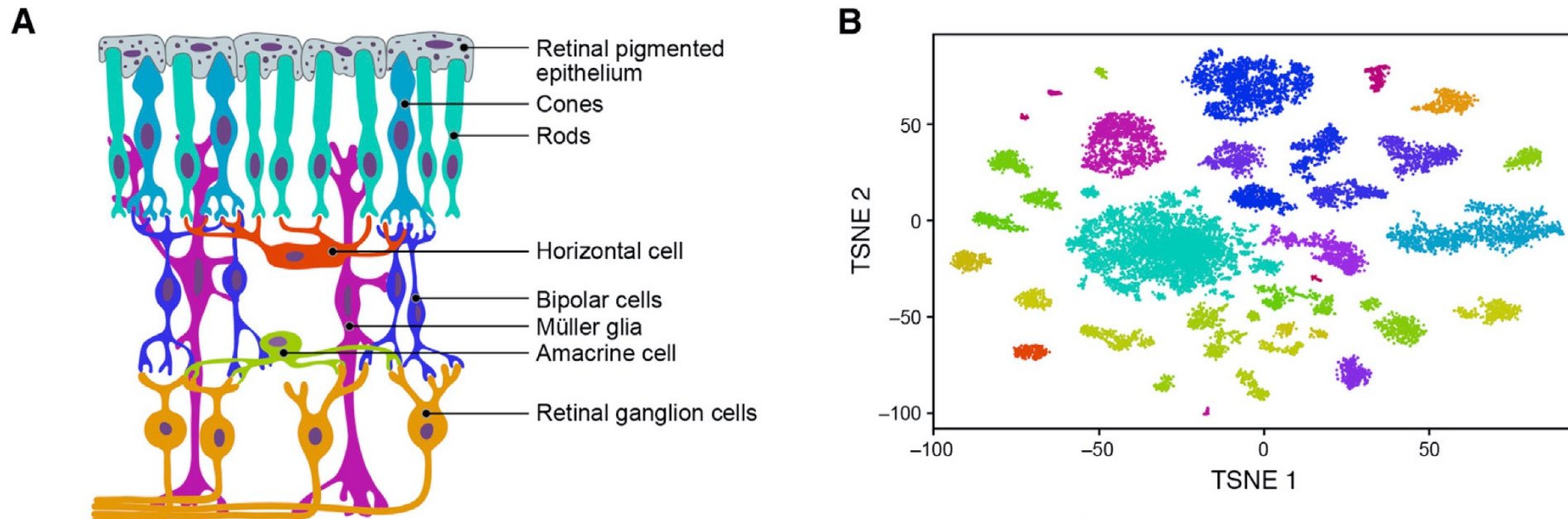
Does it make sense that PCA will be enough?

Single cell RNA sequencing

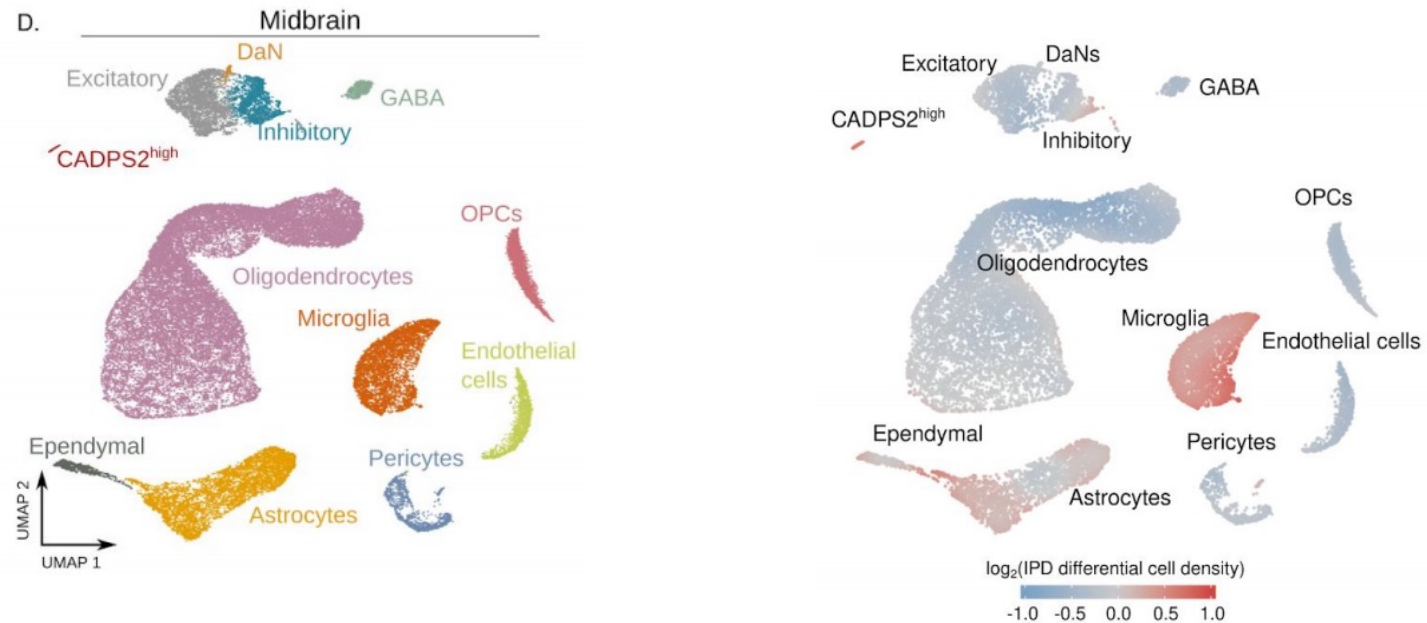


Macosko et al., Cell, 2015

- Cells visualization using non-linear dimension reduction



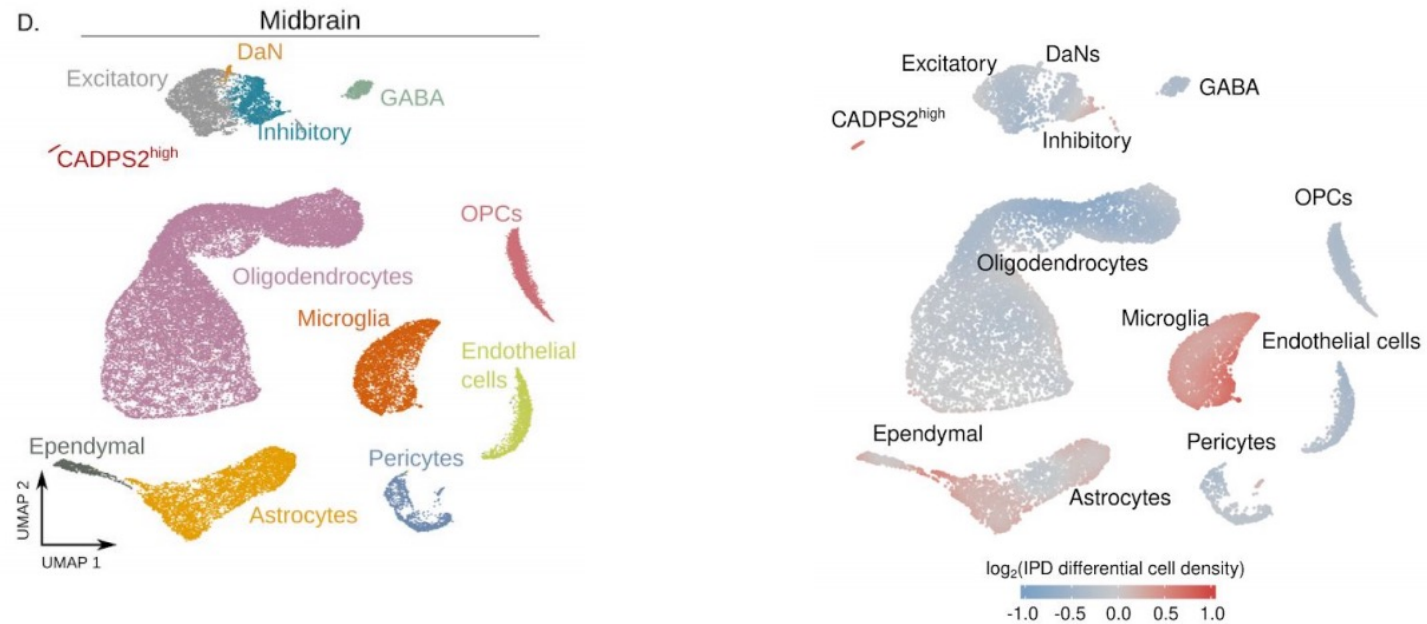
Single cell RNA sequencing in Parkinson's Disease (PD)



Smajić et al., MedRxiv, 2020

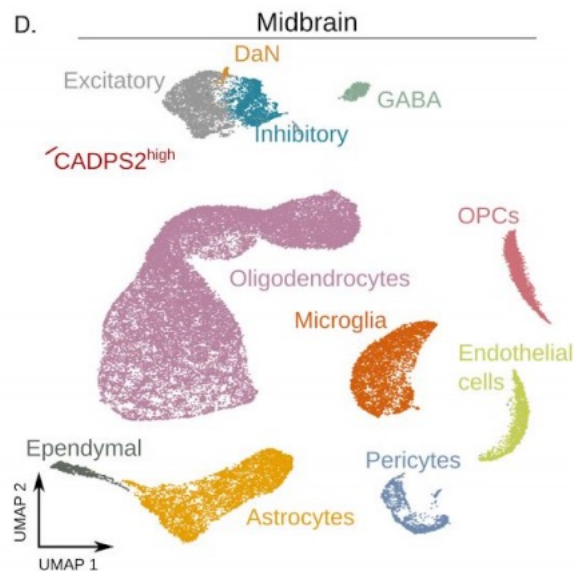
- Substantia nigra and midbrain of postmortem human donors

Is it really single cell analysis? (what happens to cells that are not in clusters?)



Smajić et al., MedRxiv, 2020

- Substantia nigra and midbrain of postmortem human donors



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If Darwin went on his voyage of the Beagle today his research would've ended with a t-SNE of the bird specimens that he collected instead of a theory of evolution

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